

5(a)	any five from: <i>Allow any temperature within the ranges given</i> 1 <u>denaturation</u> at 90–98 °C ; 2 breaks hydrogen bonds (in DNA) to, produce single strands / separate the (two) strands ; 3 <u>annealing</u> at 50–65 °C ; 4 (so that) primers, anneal / bind, (ss)DNA / complementary bases ; 5 (to provide) binding site / starting point (for Taq polymerase) ; 6 extension or elongation at 65–75 °C ; 7 Taq polymerase joins, (free) nucleotides / dNTPs, to (single) strands or Taq polymerase synthesises, complementary / new, strand (of DNA) ;	5
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Question	Answer	Marks
5(b)(i)	any four from: <i>bacteria = E.coli and virus = herpes simplex throughout</i> 1 extract / isolate / cut, DNA (sequences) using restriction enzyme or make DNA (sequences) using mRNA using reverse transcriptase or make DNA (sequences) using nucleotides ; 2 use DNA ligase to join together (the two sequences of) DNA (of bacteria and virus) ; 3 add promoter ; 4 add, <i>tTav</i> / recombinant DNA / gene, to, vector / virus / liposome ; 5 infect egg with virus or (micro)injection into egg ; A gene gun / DNA gun 6 virus adds, <i>tTav</i> / recombinant DNA / gene, into genome (of egg / <i>A. stephensi</i>) ;	4
5(b)(ii)	any one from: 1 <i>idea that</i> prevents, transcription factor, functioning / binding to promoter ; R repressor proteins / operator 2 stops RNA polymerase from binding to, DNA / promoter ;	1
5(b)(iii)	(genetically modified males) develop into adults / survive beyond the larval stage ;	1